Energía de ligadura I

Constantes

In[ ]:=
```
me=0.51099895069; (*MeV/c^2*)
mn=939.56542194; (*MeV/c^2*)
mp=938.27208943; (*MeV/c^2*)
uma=931.49410372; (*MeV/c^2*)
```

Carga de datos

In[ ]:=
```
data = Import["C:\\Users\\ruben\\Desktop\\Prácticas Nuclear\\Práctica 1 Radio nuclear y energí
data2=Rest[data];
```

In[ ]:=
```
datosAZM = Select[
   data2,
   Length[#] ≥ 7 && NumericQ[#[[1]]] && NumericQ[#[[3]]] && NumericQ[#[[7]]] &
][[All, {1, 3, 7}]];
```

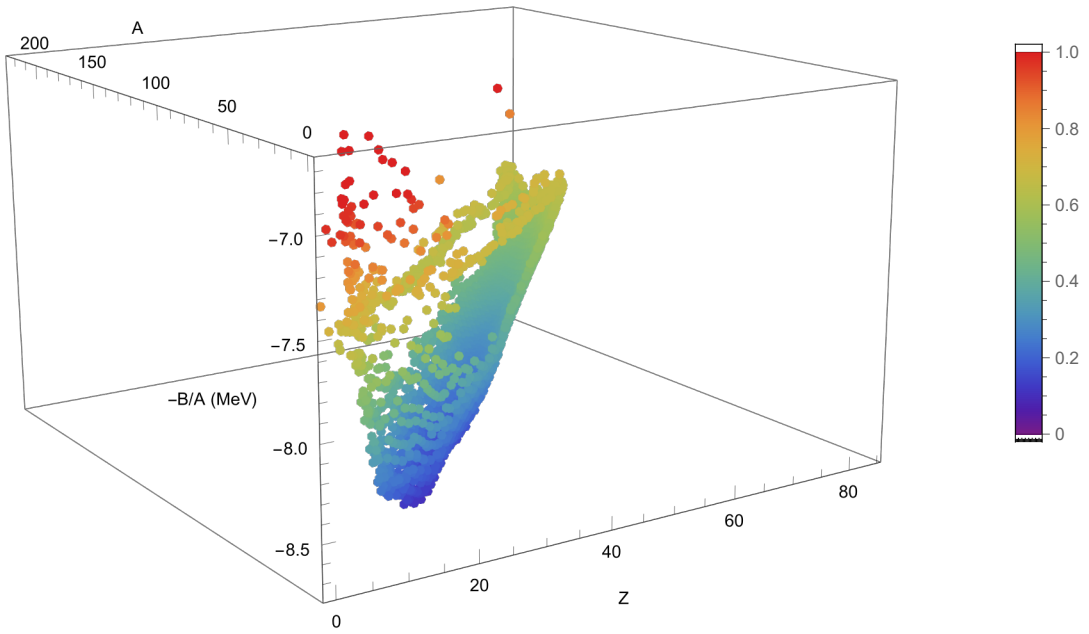
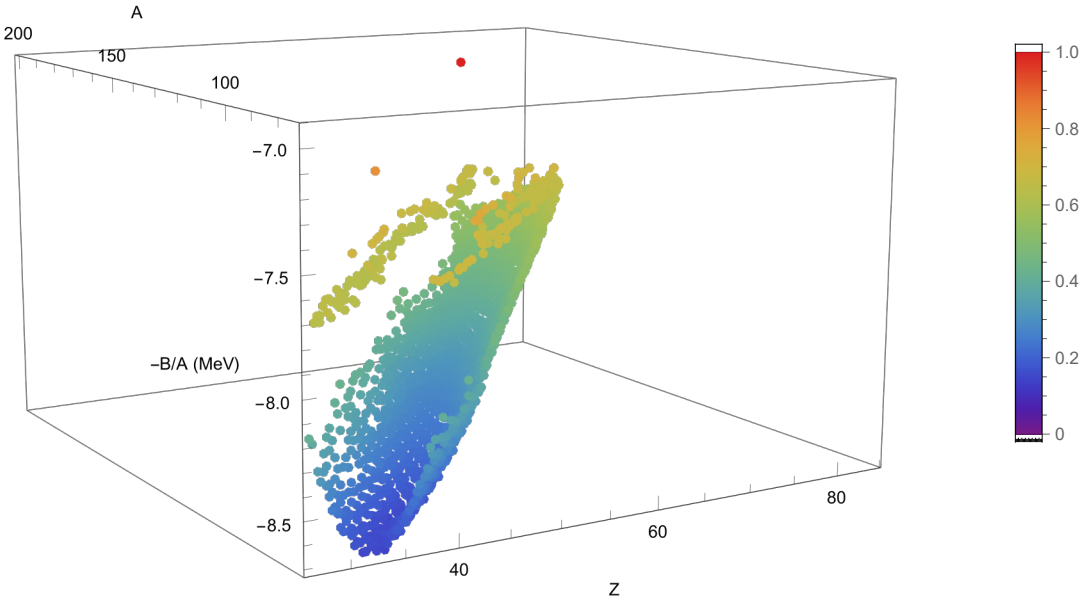
Binding Energy

In[ ]:=
```
B[A_, Z_, M_] := Z*mp + (A - Z)*mn + Z*me - (M + A*uma);
BA[A_, Z_, M_] := B[A, Z, M]/A;
```

Representación 3D

```
resultadoBA = ({#[[2]], #[[1]], -BA[#[[1]], #[[2]], #[[3]]]} &) /@ datosAZM;
ListPointPlot3D[
  resultadoBA,
  AxesLabel → {"Z", "A", "-B/A (MeV)"},
  PlotStyle → PointSize[0.01],
  ColorFunction → (ColorData["Rainbow"][Rescale[#3, {-9, -7}]] &),
  ColorFunctionScaling → False,
  BoxRatios → {1, 1, 0.6},
  ViewPoint → {2.2, -2.4, 1.4},
  ViewVertical → {0, 0, 1},
  SphericalRegion → True,
  PlotLegends → BarLegend["Rainbow"],
  ImageSize → Large
]
datosFiltrados = Select[resultadoBA, 75 ≤ #[[2]] ≤ 200 &];
ListPointPlot3D[
  datosFiltrados,
  AxesLabel → {"Z", "A", "-B/A (MeV)"},
  PlotStyle → PointSize[0.01],
  ColorFunction → (ColorData["Rainbow"][Rescale[#3, {-9, -7}]] &),
  ColorFunctionScaling → False,
  BoxRatios → {1, 1, 0.6},
  ViewPoint → {2.2, -2.4, 1.4},
  ViewVertical → {0, 0, 1},
  SphericalRegion → True,
  PlotLegends → BarLegend["Rainbow"],
  ImageSize → Large
]
```

Out[ ]=



Out[ ]=